

Implementing CI/CD Pipeline for a Microservice project

CAPSTONE PROJECT

COHORT 4 | GROUP 4

Agenda

1. About Project
2. Our Team
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4. Architecture Overview
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6. Tools selection methodology
7. Q&A

About project

This project is a comprehensive approach to strengthening a basic Continuous Integration (CI) and Continuous Deployment (CD) pipeline in the context of transitioning from a monolithic architecture to a microservices-based architecture. The project focuses on enhancing the CI/CD pipeline to meet the specific demands of microservices environments by integrating advanced code quality checks, automated testing, and infrastructure management.

OUR TEAM



Santhosh P



Sagar

Ajay

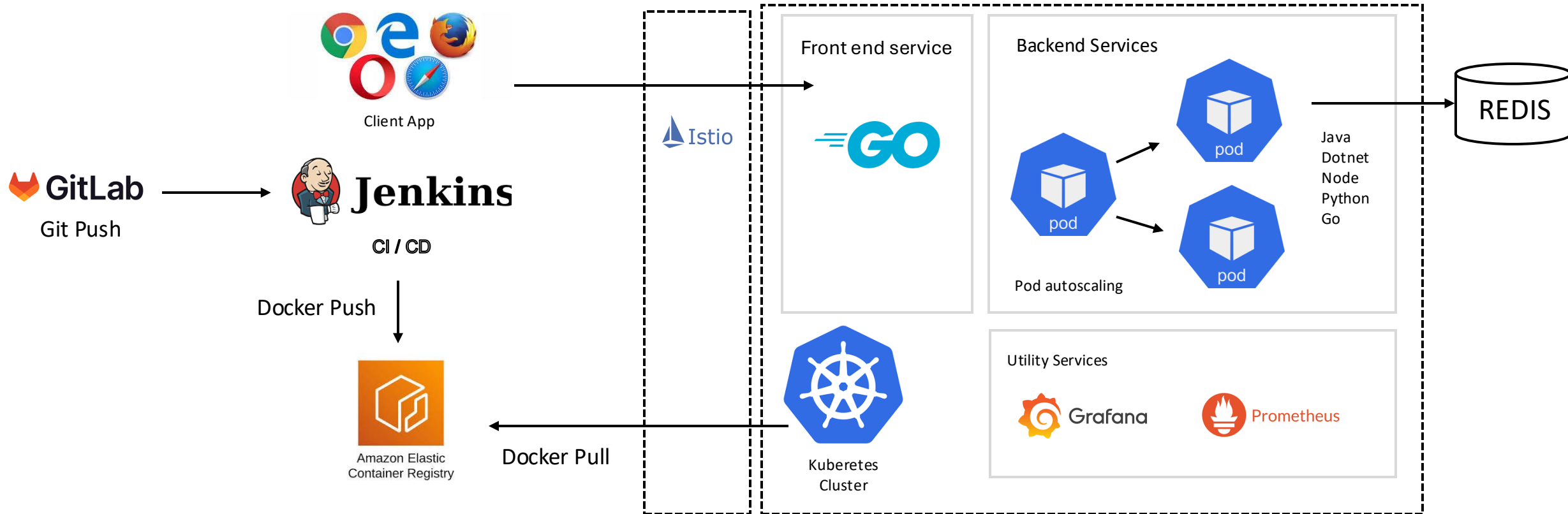


Yinebeb

Tools and technologies



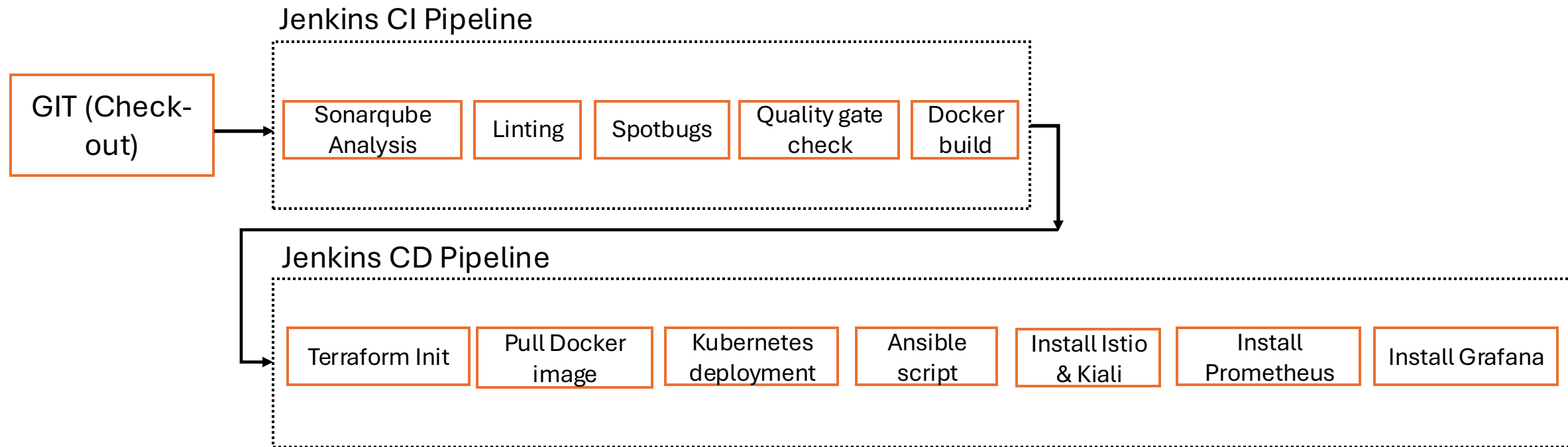
Architecture overview



Note:

* Deployment pipeline is triggered manually

Implementation details



Tool Selection Methodology

Evaluation Criteria

- Scalability & Performance
- Integration Capabilities
- Community Support
- Cost-Effectiveness
- Security Features
- Learning Curve

Selection process

- Requirement analysis
- Market research
- Proof of concept
- Team feedback

Tool	Primary Function	Key Decision Factors
Kubernetes	Container Orchestration	Scalability, Self-healing, Industry standard
Docker	Containerization	Portability, Consistency, Wide adoption
Istio	Service Mesh	Traffic management, Security, Observability
Jenkins	CI/CD Automation	Flexibility, Plugin ecosystem, Open-source
SonarQube	Static Code Quality Analysis	Comprehensive analysis, Language support, Reporting
Ansible	Configuration Management	YAML-based, Multi-platform support
Grafana	Visualization	Customizable dashboards, integrations
Prometheus	Monitoring & Alerting	Powerful querying, Active community
GitLab	Version Control	Integrated platform, Collaboration, Easy to use

Tool Selection Methodology Cont..

Decision Matrix

Tool	Scalability	Integration	Community	Cost	Security	Total
Kubernetes	5	4	5	3	4	21
Docker	4	5	5	4	4	22
Istio	4	5	5	5	3	22
Jenkins	5	5	4	4	3	21
SonarQube	4	5	5	3	4	21
Ansible	4	4	3	4	4	19
Grafana	4	4	4	5	4	21
Prometheus	5	4	3	4	4	20
GitLab	5	5	5	4	5	24

Q & A